



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fourth Semester – Fall 2023

Paper: Database Systems

Course Code: CC-215

Roll No.

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (6x5=30)

- Briefly discuss different integrity constraints used in relational databases.
- What is data dependence and what are its disadvantages?
- What is composite key and what is the concept of minimality? Briefly explain with the help of an appropriate example.
- What are ACID properties of a transaction?
- Briefly explain two-phase locking in concurrent transactions.
- Briefly explain DDL, DML, and DCL statements of SQL with the help of an example each.

Answer the following questions: (3x10=30)

Q2- Identify entities and draw ER Diagram for the following situation.

- The firm has a number of sales offices in several states. Attributes of sales office include Office Number (identifier) and Location.
- Each sales office is assigned one or more employees. Attributes of employee include Employee ID (identifier) and Employee Name. An employee must be assigned to only one sales office.
- For each sales office, there is always one employee assigned to manage that office. An employee may manage only the sales office to which he or she is assigned.
- The firm lists property for sale. Attributes of property include Property ID (identifier) and Location. Components of Location include Address, City, State, and Zip Code.
- Each unit of property must be listed with one (and only one) of the sales offices. A sales office may have any number of properties listed or may have no properties listed.
- Each unit of property has one or more owners. Attributes of owners are Owner ID (identifier) and Owner Name. An owner may own one or more units of property. An attribute of the relationship between property and owner is Percent Owned.

Q.3 Normalize the following Patient Medication Form up to 3NF. You must specify the definition, functional dependencies and anomalies at each stage.

Visit No.	VisitDate	PatNo	PatAge	PatCity	PhyNo	PhySpecialty	Diagnosis
V10020	1/13/2023	P1	35	LAHORE	D1	INTERNIST	EAR INFECTION
V10020	1/13/2023	P1	35	LAHORE	D2	NURSE PRACTIONER	INFLUENZZ
V93030	1/20/2023	P3	17	SIALKOT	D2	NURSE PRACTIONER	PREGNANCY
V82110	1/18/2023	P2	60	KASUR	D3	CARDIOLOGIST	MURMUR

Q.4 Write SQL queries using the following set of relations (tables):

EMP (EMPNO, ENAME, JOB, SAL, HIREDATE, COMM, MGR, DEPTNO)

DEPT (DEPTNO, DNAME, LOC)

- List EMPNO, ENAME, JOB, SAL, and DNAME for all the employees hired in the last five years.
- Show the name of employees and their COMM (Show 'No Commission' for those employees whose COMM is NULL) working in ACCOUNTING or SALES department.
- Find the name of the most senior employee.
- Display a list of all employees with an identifier (code) which is composite of the first two characters of their JOB and the YEAR of their HIREDATE.
- Display department wise count of employees, and average salary of the employees.